MIL-T-43233B
29 November 1973
SUPERSEDING
MIL-T-43233A
30 July 1969

MILITARY SPECIFICATION

TYING MACHINES, WIRE, HAND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers manually operated wire tying machines using round wire.

1.2 Classification. Wire tying machines covered by this specification shall be of the following types, as specified (see 6.2):

Type I	- Light duty (for 19 through 16 gauge round wire).
Type II	- Medium duty (for 15 through 12 gauge round wire).
Type III	- Heavy duty (for 12 through 8 gauge round wire).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood Cleated, Plywood
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	- Preservation, Method Of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM)

E 18 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race St., Philadelphia, Pennsylvania 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specifications for first article inspection and approval (see 4.3 and 6.3).

* 3.2 Standard product. The tying machine delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specifications requirements. All like items furnished on any one contract, including parts and subassemblies thereof shall be new and interchangeable.

* 3.3 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacture in his standard commercial tying machine provided that the completed item complies with all provisions of this specifications.

3.3.1 Metals. All metals, as applicable, including bolts, nuts, washers, capscrews, and pins shall have a corrosion-resistant finish in accordance with industry practice.

3.4 Design and construction. The tying machine shall be lever operated and have mechanism to tension, twist-seal, and trim wire strapping dispensed directly from a continuous coil. The machine shall have one or more levers as applicable. All parts subject to wear, breakage, or distortion shall be easily accessible for replacement, adjuctment, or repairs.

3.4.1 Blade. The cutting blade shall be carbon steel, and have a Rockwell C hardness number of between 55 and 62 when tested as specified in 4.4.2.1.

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3.4.2 Levers. The levers of the wire tying machine shall be fabricated of high grade steel forgings, malleable iron castings, steel castings, or aluminum or aluminum alloy forgings in accordance with industry practice. The levers shall be sufficient length and strength to afford the required leverage. The levers shall be provided with handgrips formed either integral with the levers or fitted with molded grips of oil-resistant rubber, or other suitable material.

3.4.3 Bolts, nuts, capscrews, and pins. Bolts, nuts, capscrews, and pins used in the fabrication of the wire tying machine shall be of high grade steel, properly heat treated to resist wear and shear. Means shall be provided to preclude the possibility of nuts, capscrews, or pins becoming loose under normal service conditions.

3.4.4 Type I. The type I machine, for use with 19 gauge to 16 gauge (0.0410 to 0.0625 inch diameter) wire, shall have a one-lever operating mechanism to apply to required tension, twist-seal and trim the wire over the package with one stroke of the lever. At the option of the supplier the machine may have two (2) levers; one lever to apply tension to the wire and the other to twist-seal and trim the wire. The machine shall not fail when tested as specified in 4.5.1.1.

3.4.5 Type II. The type II wire tying machine shall be designed with two levers, one to apply a desired tension to 15 gauge through 12 gauge (0.0720 to 0.1055 inch diameter) round wire with 2 to 3 strokes of the tension lever, and the other to twist-seal and trim the wire with 3 strokes of the twist-seal and trim lever. The machine shall apply tension, twist-seal, and trim the wire when tested as specified in 4.5.1.2 or 4.5.1.3.

3.4.6 Type III. The type III wire tying machine shall be designed with three levers, one to apply a desired tension to 12 gauges through 8 gauge (0.1055 to 0.1620) inch diamter) wire with 2 to 3 strokes of the tension lever, the second to twist the wire with nine strokes of the twist-seal lever, and the third to trim the wire with one stroke of the cutoff lever. The machine shall apply tension, twist-seal, and trim the wire when tested as specified in 4.5.1.2 and 4.5.1.3.

3.5 Finish. The machine shall be finished in accordance with industry practice, and be free from burrs, sharp edges, or projections.

3.6 Marking for identification. The wire tying machine shall be marked in a permanent manner with manufacturer's identification in accordance with MIL-STD-130.

3.7 Workmanship. The machine levers shall be free from flash and irregular shapes. Screws, pins, bolts, and nuts shall not become loose upon assembly, and pins shall not wear or shear. Welds shall be free of burn holes, cracks, fractures, incomplete fusion, or scale or flux deposits.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article. When a first article is required, the tying machine shall be examined/or tested as specified in 4.5.1.1 through 4.5.1.3. The presence of any defect or failure of any test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specificaitons, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specificaiton or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 Intermediate testing. The blade test shall conform to ASTM E 18 to determine compliance with 3.4.1. The sample unit shall be one blade. The lot shall be all the blades submitted for inspection at one time. Three determinations shall be taken on each sample unit and the results shall be the average of the three determinations. The inspection level shall be S-2 with an acceptable quality lever (AQL) of 2.5 defects expressed in terms of defects per hundred units.

* 4.4.3 End item inspection. The inspection lot shall be all tying machines of one type offered for inspection at one time.

4.4.3.1 Visual inspection. The tying machine shall be examined for the defects in table I. The inspection level shall be II with an AQL (acceptable quality level) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not as specified		X
	Burrs, sharp edges, or projections	X	
Design, construction and workmanship	Not in accordance with the specified requirements	X	
	Part missing or not specified type	X	
	Fractured, split, punctured, dented, deteriorated, bowed, malformed or otherwise impaired	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Design, construction, and workmanship (con't)	Misplaced, not in proper alignment	X	
	Part loose		X
	Functioning component that requires abnormal force to operate		X
	Adjustable assembly that cannot be adjusted	X	
	Component not readily accessible for servicing where required		X
Welding	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed		X
Bolts, nuts, screws, or pins	Missing, broken, stripped, fractured or loose	X	
Marking indentification	Missing, incomplete, not legible		X

* 4.4.3.2 Dimensional examination. The tying machine shall be examined for compliance with dimensions specified. Any dimension not within this specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

* 4.4.3.3 End item testing. When a first article is not required, the first produced tying machine shall be tested as specified in 4.4.3. Each sample tying machine shall be tested as specified in 4.5.1.1 through 4.5.1.3. Each sample to pass a test shall be cause for rejection of the inspection lot or failure to pass a test shall be a defect. The inspeciton level shall be S-2 with an AQL of 2.5 defects, expressed as defects per hundred units.

4.4.4 Examination of preparation for delivery requirements. An examination shall be made to determine that preservation, packaging, packing, and marking requirements of section 5 are complied with. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery, with the exception that it need not be sealed. The lot shall be the number of shipping containers offered for delivery at one time. The inspection level shall be S-3 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

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TABLE II. Examination of packing and marking

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location sequence, or method of application
Cleaning	Not cleaned as specified
Preservation	Preservative improperly applied or missing
Materials	Any component missing Any component damaged
Workmanship	Inadequate application of components, such as: incomplete blocking, cushioning, closure of container flaps, when applicable, inadequate stapling and strapping Handles, when applicable, not secured
Weight	Exceeds specified net weight
Content	More or less than required

4.5 Tests

4.5.1 Performance tests.

4.5.1.1 Type I. Using the wire directly from a continuous coil, drape the wire (16 gauge) around a package and through machine to the tensioning drum or holding gripper mechanism, and then operating the one or two lever machines, tension, twist-seal, and trim to determine conformance with 3.4.4.

4.5.1.2 Type II. Using the wire directly from a continuous coil, drape the wire (12gauge) around a package and through machine to tensioning drum or holding the gripper mechanism. Apply 2 to 3 strokes to the tensioning lever and a maximum of three strokes of the twist-seal and trim lever to determine conformance with 3.4.5 or 3.4.6.

4.5.1.3 Type III. Using wire directly from a continuous coil, drape the wire (8 gauge) around a package and through machine to tensioning drum or holding gripper mechanism. Apply 2 to 3 strokes to the tension lever and a maximum of nine strokes of the twist-seal lever and actuate the trim lever to determine conformance with 3.4.5 or 3.4.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each machine shall be cleaned process C-1, dried, and preserved Method I of MIL-P-116.

* 5.1.1.1 Packaging. Each machine, preserved as specified in 5.1.1 shall be packed in a snug fitting fiberboard box conforming to type CF of class domestic, grade open, style RSC of PPP-B-636.

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Each machine shall be immobilized within the box with scored and folded or die-cut pads made from the same material as the box. The closure shall be in accordance with method I of the appendix PPP-B-636. When specified (see 6.2), boxes shall be placed in a minimum 3 mils bag and the bag heat scaled.

5.1.2 Level C. Each machine shall be preserved and packaged to afford adequate protection against deterioration or damage during known favorable condition of handling and storage from the supply source to receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Machines of one type only, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class weather resistant, grade V2s, style RSC of PPP-B-636 or overseas type, style A, B, or I of PPP-B-601. The weight of contents of the shipping container shall not exceed 65 pounds. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Unless otherwise specified (see 5.2.2.1) machines of one type only preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF, class domestic, minimum grade 275, style RSC of PPP-B-636. The weight of contents of the shipping container shall not exceed 65 pounds. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be made of grade V3c, V3s, or V4s fiberboard. The box shall be fabricated in accordance with PPP-B-636 and closed in accordance with the weather resistant requirements of the appendix of PPP-B-636.

5.2.3 Level C. Machines, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transprotation rate for such supplies. Containers shall be in accordance with National Motor Freight Classification Rules or Uniform Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers, consolidators or pallet loads, shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The wire tying machines covered by this specification are intended for tensioning, twist-scaling, and trimming round steel wire strapping in packaging operations.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of tying machine required (see 1.2).
(c) First article. When a first article is required (see 3.1, 4.3 and 6.3).
(d) Selection of the applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

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(e) When weather-resistant wrap is required (see 5.1.1.1).
(f) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

* 6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. the first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

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Review activities:

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